Smart Factory Resource Allocation System (SRAS)

Sample Database Data for Learning and Presentation

1. Factory Settings

Field	Value
Workspace Name	Smart Factory Resource Allocation (SRAS)
Plant Location	Industrial Corridor Sector 7, Plant 01
Time Zone	Asia/Kolkata
Max Employee Workload %	85
Prioritize Skill Match	True
Reserve Reorder Stock	True
Conflict Notifications	True
Low Stock Notifications	True

2. Employees

ID	Name	Employee No	Skills	Shift	Workload %
emp_rajesh	Rajesh Sharma	EMP-101	CNC, Precision Milling, G-Code Optimization	Shift A	45
emp_priya	Priya Patel	EMP-102	Assembly, Quality Control, ISO Inspection	Shift A	60
emp_amit	Amit Verma	EMP-103	MIG/TIG Welding, Sheet Metal, Fabrication	Shift B	30
emp_sneha	Sneha Kulkarni	EMP-104	CNC, Robotics, Tool Setting	Shift B	55

3. Machines

ID	Machine	Type	Location	Compatible Materials
mach_cnc_01	Haas 5-Axis VMC Center	CNC	Bay 1 - High Precision	Steel & Alloys, Aluminum, Brass & Bronze
mach_lathe_02	Mazak CNC Heavy Lathe	Lathe	Bay 1 - Turning	Steel & Alloys, Brass & Bronze
mach_weld_03	Panasonic Robotic Arc Welder	Welder	Bay 2 - Fabrication	Steel & Alloys, Sheet Metal
mach_assm_04	Vision Automated Assembly Cell	Assembly	Bay 3 - Packaging	Polymers, Steel & Alloys

4. Materials

SKU	Name	Category	Qty On Hand	Qty Reserved	Reorder Point
MAT-STL-316	316L Stainless Steel Rods	Steel & Alloys	550	45	100
MAT-ALU-606	6061-T6 Aluminum Billets	Aluminum	380	30	80
MAT-BRS-360	C360 High-Tensile Brass Bar	Brass & Bronze	190	0	40
MAT-SHT-GAL	Galvanized Steel Sheet 2mm	Sheet Metal	220	25	50
MAT-POL-001	High-Impact Polycarbonate Resin	Polymers	420	0	90

5. Production Tasks

Task ID	Task #	Title	Status	Machine Type	Required Material
task_valve_body_01	101	Cryogenic Flow Control Valve Run	in_progress	CNC	45 kg 316L Steel
task_flange_02	102	Aerospace Bracket Flange Batch	scheduled	CNC	30 kg 6061 Aluminum

Note: This PDF contains the sample dataset used by the SRAS project for demo, learning, and viva presentation.